EXERCISE 8.1

Sequences

Questions with Step-by-Step Solutions

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Question 1

Find the first five terms of the sequence in which the nth term is given by

(i) $t_n = 3n - 4$,

(ii) $t_n = 2n - 5$,

(iii) $t_n = 2n + 2$,

for $n \geq 1$.

Solution

(i) Given:

$$t_n = 3n - 4$$

Put $n = 1, 2, 3, 4, 5$.

$$\begin{aligned} t_1 &= 3(1) - 4 = -1, \\ t_2 &= 3(2) - 4 = 2, \\ t_3 &= 3(3) - 4 = 5, \\ t_4 &= 3(4) - 4 = 8, \\ t_5 &= 3(5) - 4 = 11. \end{aligned}$$

Therefore,

$$\boxed{-1,\ 2,\ 5,\ 8,\ 11}$$

(ii) Given:

$$t_n = 2n - 5$$

$$\begin{aligned} t_1 &= 2(1) - 5 = -3, \\ t_2 &= 2(2) - 5 = -1, \\ t_3 &= 2(3) - 5 = 1, \\ t_4 &= 2(4) - 5 = 3, \\ t_5 &= 2(5) - 5 = 5. \end{aligned}$$

Therefore,

$$\boxed{-3,\ -1,\ 1,\ 3,\ 5}$$

(iii) Given:

$$t_n = 2n + 2$$

$$\begin{aligned} t_1 &= 2(1) + 2 = 4, \\ t_2 &= 2(2) + 2 = 6, \\ t_3 &= 2(3) + 2 = 8, \\ t_4 &= 2(4) + 2 = 10, \\ t_5 &= 2(5) + 2 = 12. \end{aligned}$$

Therefore,

Question 2

Find the 10th and 15th terms of the sequence

$$t_n = 5n - 3, \qquad n \geq 1.$$

Solution

Given,

$$t_n = 5n - 3$$

For the 10th term, put $n = 10$:

$$\begin{aligned} t_{10} &= 5(10) - 3 \\ &= 50 - 3 \\ &= \boxed{47} \end{aligned}$$

For the 15th term, put $n = 15$:

$$\begin{aligned} t_{15} &= 5(15) - 3 \\ &= 75 - 3 \\ &= \boxed{72} \end{aligned}$$

Final Answer

$$\boxed{t_{10} = 47, \qquad t_{15} = 72}$$

Question 3

Determine whether 97 and 172 are terms of the sequence

$$t_n = 5n - 3, \qquad n \geq 1.$$

Solution

For 97 to be a term of the sequence,

$$\begin{aligned} 5n - 3 &= 97 \\ 5n &= 100 \\ n &= 20. \end{aligned}$$

Since $n = 20$ is a positive integer, 97 is a term of the sequence.

$$\boxed{97 = t_{20}}$$

Now check 172:

$$\begin{aligned} 5n - 3 &= 172 \\ 5n &= 175 \\ n &= 35. \end{aligned}$$

Since $n = 35$ is a positive integer, 172 is also a term of the sequence.

$$\boxed{172 = t_{35}}$$

Final Answer

$$\boxed{\text{97 and 172 are both terms of the sequence.}}$$

Question 4

Which term of the sequence

$$t_n = 5n - 3, \qquad n \geq 1$$

is 607?

Solution

We need to find n such that

$$t_n = 607.$$

Given,

$$\begin{aligned} 5n - 3 &= 607 \\ 5n &= 610 \\ n &= 122. \end{aligned}$$

Therefore, 607 is the 122nd term of the sequence.

Final Answer

$$\boxed{607 = t_{122}}$$

Question 5

A sequence is given by the recursive rule

$$t_1 = 5, \qquad t_{n+1} = t_n + 3, \qquad n \geq 1.$$

(a) Find the first five terms of this sequence.

(b) Is 52 a term of this sequence? If so, which term is it?

Solution

We are given

$$t_1 = 5$$

and

$$t_{n+1} = t_n + 3.$$

So, each term is obtained by adding 3 to the previous term.

$$\begin{aligned} t_1 &= 5, \\ t_2 &= 5 + 3 = 8, \\ t_3 &= 8 + 3 = 11, \\ t_4 &= 11 + 3 = 14, \\ t_5 &= 14 + 3 = 17. \end{aligned}$$

Hence, the first five terms are

$$\boxed{5,\ 8,\ 11,\ 14,\ 17}$$

Now check whether 52 is a term.
The sequence is an AP with

$$a = 5, \qquad d = 3.$$

Therefore,

$$\begin{aligned} t_n &= a + (n-1)d \\ &= 5 + (n-1)3. \end{aligned}$$

For $t_n = 52$,

$$\begin{aligned} 5 + 3(n-1) &= 52 \\ 3(n-1) &= 47 \\ n - 1 &= \frac{47}{3} \\ n &= \frac{50}{3}. \end{aligned}$$

But $\frac{50}{3}$ is not a positive integer.
Therefore, 52 does not occur in the sequence.

Final Answer

$$\boxed{\text{First five terms: } 5,\ 8,\ 11,\ 14,\ 17}$$

$$\boxed{52 \text{ is not a term of the sequence.}}$$

Question 6

Let

$$T_1 = 1, \qquad T_2 = 2, \qquad T_3 = 4$$

and

$$T_n = T_{n-1} + T_{n-2} + T_{n-3}, \qquad n \geq 4.$$

Find T_4, T_5, T_6, T_7 and T_8.

Solution

We use the recursive rule

$$\boxed{T_n = T_{n-1} + T_{n-2} + T_{n-3}}$$

Given,

$$T_1 = 1, \qquad T_2 = 2, \qquad T_3 = 4.$$

Therefore,

$$\begin{aligned} T_4 &= T_3 + T_2 + T_1 \\ &= 4 + 2 + 1 \\ &= \boxed{7} \end{aligned}$$

Next,

$$\begin{aligned} T_5 &= T_4 + T_3 + T_2 \\ &= 7 + 4 + 2 \\ &= \boxed{13} \end{aligned}$$

Next,

$$\begin{aligned} T_6 &= T_5 + T_4 + T_3 \\ &= 13 + 7 + 4 \\ &= \boxed{24} \end{aligned}$$

Next,

$$\begin{aligned} T_7 &= T_6 + T_5 + T_4 \\ &= 24 + 13 + 7 \\ &= \boxed{44} \end{aligned}$$

Finally,

$$\begin{aligned} T_8 &= T_7 + T_6 + T_5 \\ &= 44 + 24 + 13 \\ &= \boxed{81} \end{aligned}$$

Final Answer

$$\boxed{T_4 = 7, \quad T_5 = 13, \quad T_6 = 24, \quad T_7 = 44, \quad T_8 = 81}$$

Key Idea

Quick Concept: How to Find a Term?

If an explicit formula is given, simply substitute the required value of n.

$$\boxed{\text{Put the value of } n \text{ in the formula}}$$

If a recursive rule is given, start from the first term and generate the next terms one by one.

$$\boxed{\text{Previous Term} \longrightarrow \text{Next Term}}$$

"Find the pattern, understand the rule, and solve!"

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